Week 6 Final Capstone Report

Sales Data Analytics & Business Intelligence

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1. Project objective

Extend the Week 5 sales dashboard into an end-to-end analytics capstone covering Git/GitHub, data preparation, exploratory data analysis, visualizations, Power BI, and actionable recommendations.

2. Dataset and cleaning

The source data includes Sales_Raw.csv, Sales_Clean.csv, Customers.csv, and Products.csv from Week 5. The raw file contains 121 rows, one repeated Sale_ID, one missing Quantity, and one missing Discount. The script removes duplicate Sale_ID values, infers missing values from the existing Sales_Amount where possible, and recalculates sales using percentage discounts.

3. KPI summary

Metric	Result
Clean transactions	120
Total sales	■9,534,180.00
Average transaction	■79,451.50
Highest transaction	■432,250.00
Lowest transaction	■1,080.00

4. Key findings

• The cleaned dataset contains 120 transactions and total sales of ■9,534,180.

• Average transaction value is ■79,451.50; the highest transaction is ■432,250 (S0032) and the lowest is ■1,080 (S0086).

• Computers is the top sales category at ■5,371,450.

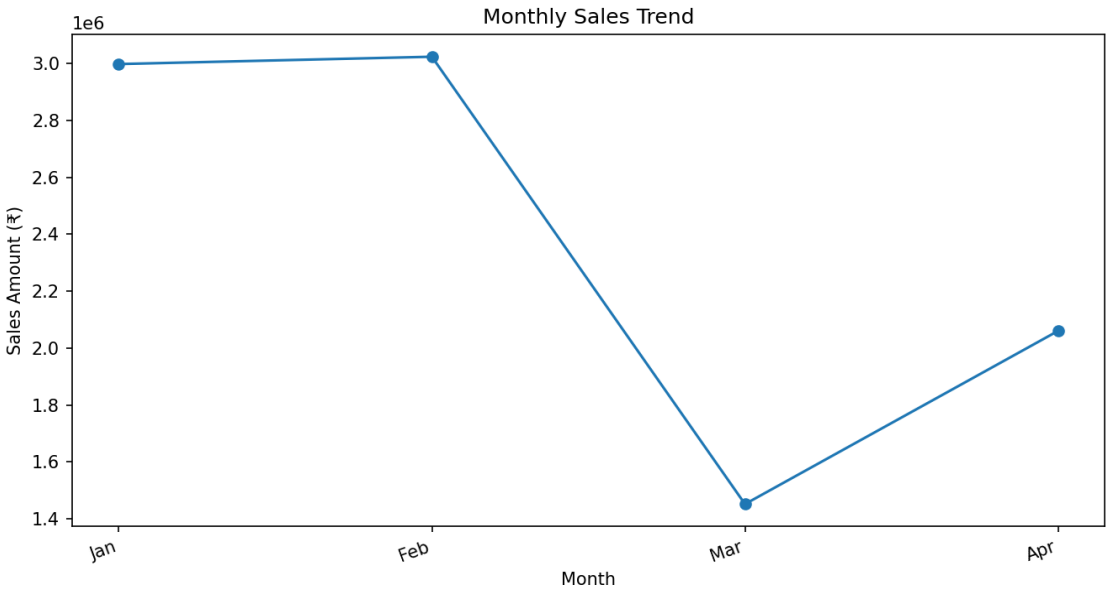
• East is the leading region at ■5,490,620.

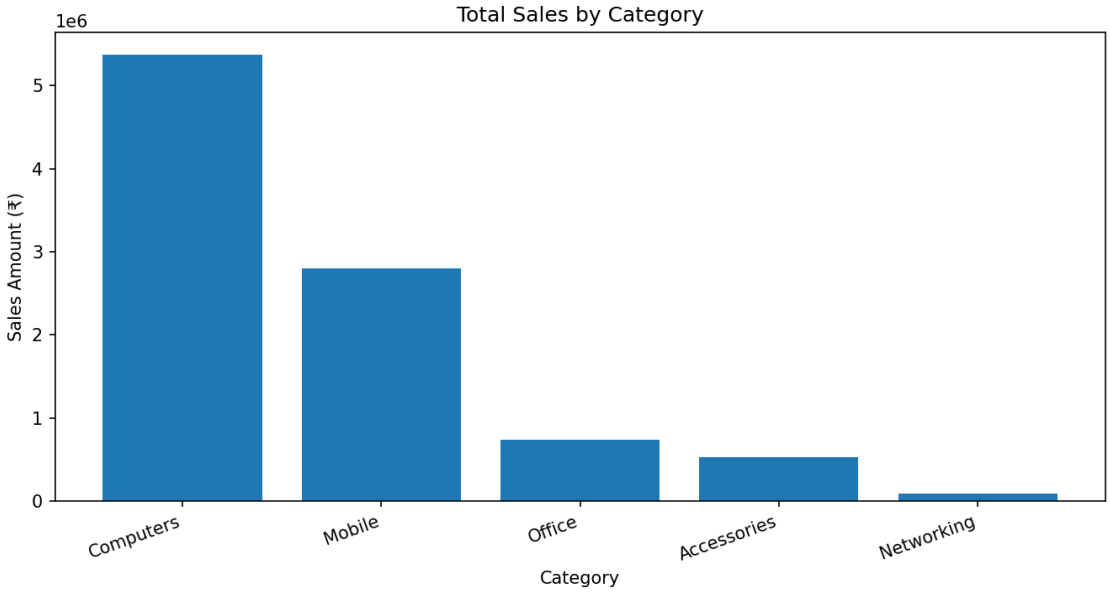
• Corporate is the largest customer segment by sales at ■4,168,450.

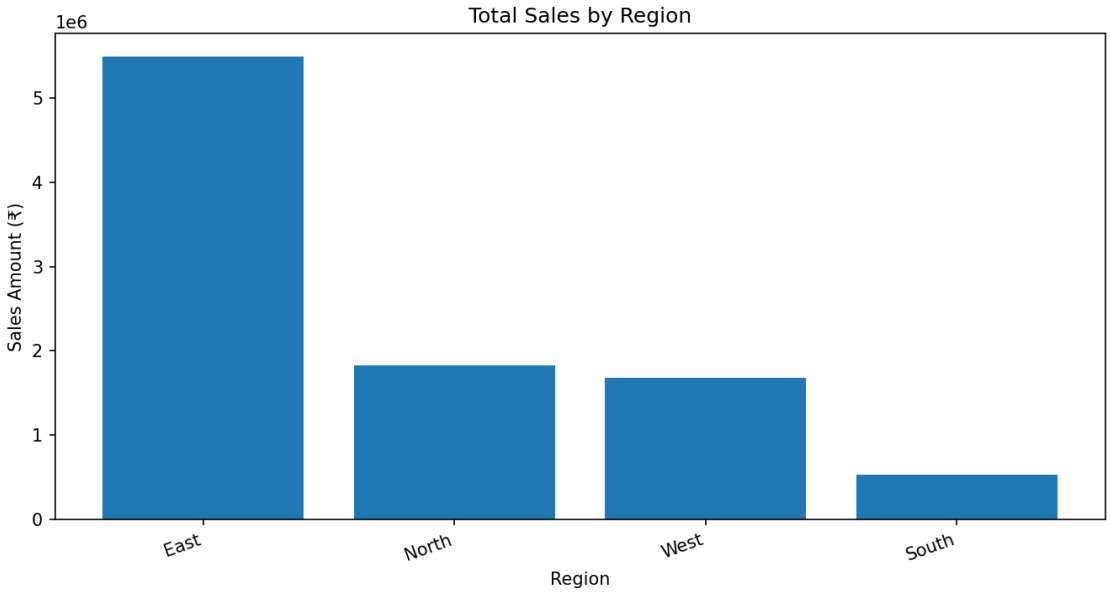
• Feb has the highest monthly sales at ■3,023,670; Mar is lowest at ■1,452,015.

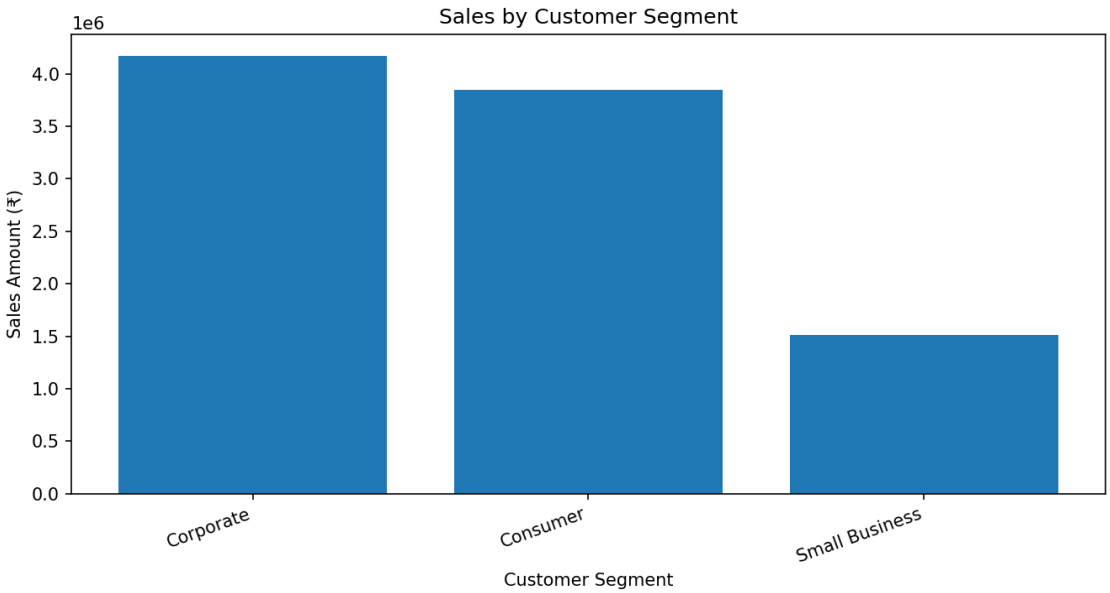
5. Visualizations

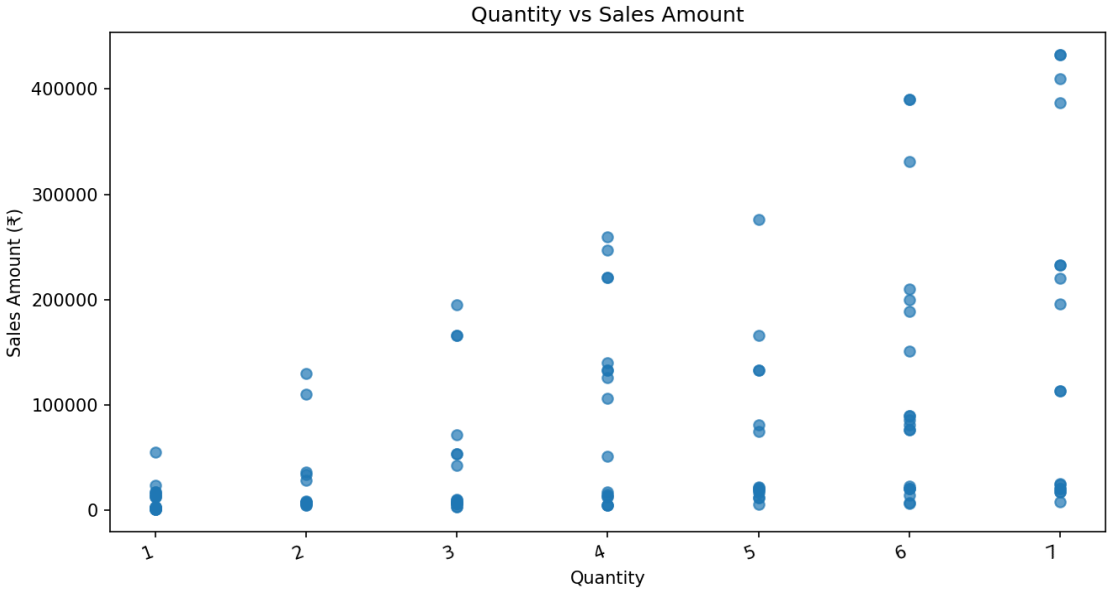
Six figures are generated in the visualizations folder: monthly trend, category comparison, regional comparison, customer segment comparison, quantity versus sales, and sales distribution.

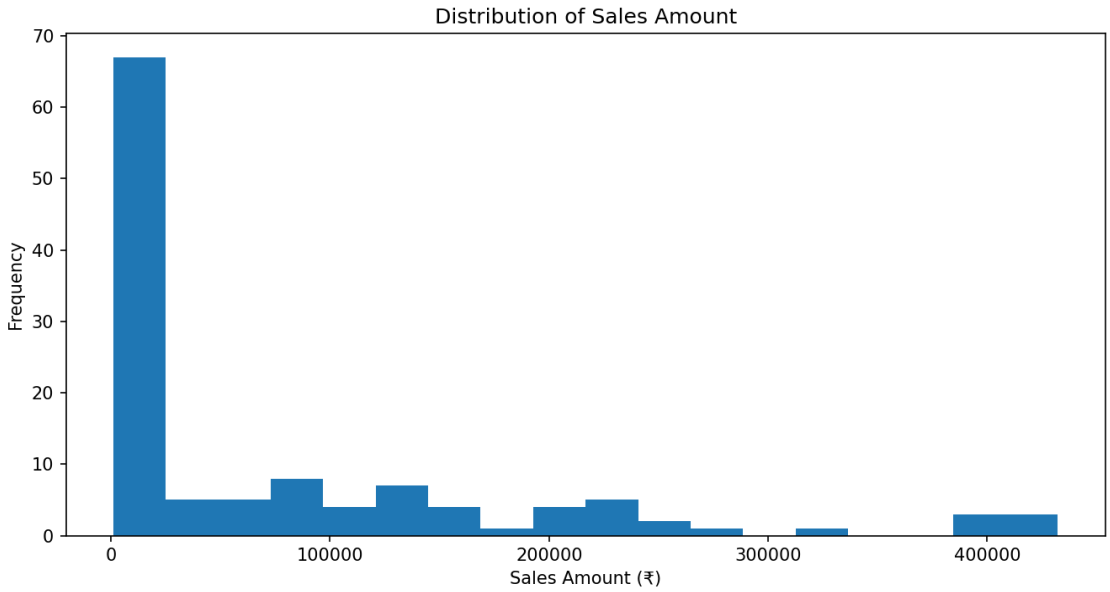

Monthly Sales Trend
1e6
3.0
2.8
2.6
2.4
2.2
2.0
1.8
1.6
1.4
Sales Amount (₹)
Jan
Feb
Mar
Apr
Month


Total Sales by Category
1e6
5
4
3
2
1
0
Sales Amount (₹)
Computers
Mobile
Office
Accessories
Networking
Category


Total Sales by Region
1e6
5
4
3
2
1
0
Sales Amount (₹)
East
North
West
South
Region


Sales by Customer Segment
1e6
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Sales Amount (₹)
Corporate
Consumer
Small Business
Customer Segment


Quantity vs Sales Amount
400000
300000
200000
100000
0
Sales Amount (₹)
1
2
3
4
5
6
7
Quantity


Distribution of Sales Amount
70
60
50
40
30
20
10
0
Frequency
0
100000
200000
300000
400000
Sales Amount (₹)

6. Business recommendations

1. Maintain stock and service capacity for the highest-sales category and region, while monitoring stock-outs and margin rather than sales alone.

2. Review the lowest-performing category, region, and customer segment; test a limited promotion or bundle and compare results against a baseline.

3. Check discount levels and transaction values regularly so high revenue is not mistaken for high profitability.

7. Power BI and GitHub

The actual Week 5 PBIX and screenshot are included in the package. For Week 6, save a separate PBIX copy if making dashboard changes. Use at least three meaningful Git commits covering datasets, analysis outputs, and final documentation/dashboard evidence.

8. Conclusion

This project links data preparation, descriptive analysis, visual storytelling, an interactive dashboard, and business recommendations. Validate each result against the generated CSV outputs before final submission.